

Sparse4D Battery CT Reconstruction

Complete project explanation and technical record

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4 October 2026

This project turns X-ray views into a 3D estimate of a battery interior, tests a learned correction on electrode simulations, and provides interactive screens for explaining and comparing the results. It is complete as a working research demonstration. Connection to a live scanner and calibrated industrial inspection remain unfinished.

The main findings

Experiment	Finding	Scope
NMC training	37.64% lower hybrid voxel RMSE vs SIRT23	Six regions within one scan
Frozen 2000bar transfer	35.38% lower hybrid voxel RMSE vs SIRT23	Separate stack; specimen independence unknown
Measured Waygate NMC replay	5.11% higher hybrid fit projection RMSE vs SIRT23	Observed negative measured transfer
Waygate physics diagnostic	300-view FDK: 0.271 s; 600-view: 0.358 s	256 cubed; solver timing only

Reading guide: Start with the workflow and glossary, then read the data and methods, results, interfaces, challenges, and real-time roadmap. The final pages provide provenance and numbered references. External papers explain methods; project-specific results come from the saved experiment records [10].

Keywords: Sparse-view computed tomography, cone-beam reconstruction, battery imaging, FDK, SIRT, residual 3D U-Net, GPU acceleration, simulated projections, domain shift, data consistency, reproducibility, interactive visualization.

1 Project purpose and what it does

Imagine looking at a battery from one direction with X-rays. That view combines material along each ray. It does not directly show a clean interior slice. Taking views from many angles gives a reconstruction algorithm enough information to estimate a 3D volume. A slice viewer then shows selected sections of that estimate.

The project asks two related questions. First, can a small learned correction improve reconstruction from sparse simulated NMC views? Second, how do view count and voxel resolution affect the reconstruction of real archived battery projections? These questions have different evidence requirements and are reported separately.

Current capabilities

The software loads TIFF stacks or prepared projection arrays, creates controlled simulation inputs, runs GPU reconstruction, optionally applies a frozen model, reports numerical errors and stage timings, and exports review files. The interfaces let a presenter show differences rather than relying only on summary percentages.

Meaning of the project name

Sparse4D is the established project name. The battery experiments in this report are static 3D reconstructions. The fourth dimension would be time, requiring a sequence of volumes and a separate motion/temporal evaluation. Neither repeated noise seeds nor viewing a volume from different directions makes the present experiment 4D.

Useful applications and their limits

The prototype supports education, reconstruction research, inspection workflow design, and engineering comparisons. A future validated system could assist inspection of battery internal structure. The present version has no defect labels, no automatic defect detector, no state-of-health predictor, and no production acceptance rule. Seeing clearer layers does not prove that a safety-critical defect is preserved.

Term	Plain meaning
Projection	A measured or simulated 2D X-ray view from one angle
Voxel	A small cell in a reconstructed 3D volume
FDK	A fast circular cone-beam reconstruction method [1]
SIRT	An iterative method that fits projections [3]
Residual model	A network that adds a learned correction to an initial image
Domain shift	A change between training conditions and deployment data

2 System workflow and architecture

The system has a data layer, a reconstruction layer, and a presentation layer. Separating them makes results reproducible and helps prevent the interface from presenting a simulation as a scanner measurement.

Stage	Input and operation	Output
Data preparation	Read TIFFs, verify shape and ordering, preserve metadata and hashes	Volume crop or projection array
Simulation branch	Scale a crop and forward-project it with assumed geometry	Synthetic line integrals
Measured branch	Select logged angles and apply explicitly provisional preprocessing	Archived line integrals
Physics reconstruction	FDK followed by SIRT where specified	Baseline 3D volume
Optional learned correction	Frozen residual 3D U-Net plus optional three SIRT steps	Raw and hybrid volumes
Evaluation	Calculate the appropriate errors and record timing boundaries	JSON report and review arrays
Presentation	Render slices, comparisons and explanatory controls	Offline review or local inference UI

Physics and AI play different roles

FDK and SIRT use a forward model of X-ray geometry. The neural model learns the remaining difference between a simulated reconstruction and its reference. A learned correction can help in its training domain and fail after a substantial change in object type, scale or preprocessing. The Waygate negative transfer demonstrates why a physics baseline must remain visible.

Local app and offline review

The local GPU app performs computation on the PC and provides progress and result downloads. The standalone HTML review embeds saved image arrays and needs no GPU or server. Selecting 300 views in that review changes the displayed completed result; it does not reconstruct a new scan or alter scanner acquisition.

Artifacts

The saved records include `best_model.pt`, `protocol.json`, `training_history.json`, `experiment report.json` files, `review_slices.npz` files and comparison PNGs. Review ZIPs package evidence for inspection. The fixed checkpoint hash is a stronger identifier than a filename, because unrelated models can share the same filename.

3 Datasets and data preparation

Two kinds of data are used. The NMC data are already reconstructed grayscale electrode slices from the Battery 3D Images collection [8]. The Waygate data include acquired X-ray projections and a vendor reconstruction of the top of an 18650 cylindrical battery [9]. The electrode stacks are not measured projection inputs.

Data	Verified local structure	Use
NMC 0bar	218 unique TIFFs, 2048 x 2048, uint16	Training and spatial validation/test
NMC 2000bar	218 unique TIFFs, 2048 x 2048, uint16	Frozen separate-stack and noise checks
Waygate projections	1201 TIFFs, 750 x 750, uint16	1200 used views; repeated endpoint excluded
Waygate reference	750 cubed vendor volume; supplied middle TIFF	Visual context only
Calibration files	Bright 8000 x 4000; dark 4000 x 4000	Mapping/correction status unresolved

Duplicate archives

NMC archives initially contained 436 entries but only 218 unique paths per examined stack. Extraction counted unique TIFF files rather than treating duplicate entries as extra scans. Individual slice checks confirmed shape, dtype and single-page TIFF structure. This step prevents inflated dataset counts.

Metadata files

The PCA file provides geometry and acquisition settings; the PCP file provides image numbers, angles and timestamps; the PCR file describes vendor reconstruction settings. The DTXML file contains acquisition context. A metadata file is not a reconstructed image. The uploaded recon_375 TIFF, rather than its PCR settings, provided the visual reference.

Correct input interpretation

A stack of reconstruction TIFFs can serve as a simulation reference or an enhancement input. Measured reconstruction requires raw/exported projections plus compatible geometry and confirmed preprocessing. Uploading a metadata file alone cannot produce a volume. Applying dark/flat subtraction to data already corrected by a scanner could introduce a second, incorrect correction.

4 Spatial split and simulation protocol

The NMC pilot uses one fixed arrangement of 128-cubed regions. Twenty-four are training regions, six are validation regions, and six are test regions. All occupy the same central 128-slice Z slab, with non-overlapping XY positions and gaps between split strips.

Split	X origins	Y origins	Z interval
Training	256, 448, 640, 832	256, 448, 640, 832, 1024, 1216	[45,173)
Validation	1408	Same six origins	[45,173)
Test	1728	Same six origins	[45,173)

Intervals are zero-based with an exclusive upper bound. A prior central crop at X and Y [960,1088) was excluded because it had already been evaluated. The split avoids exact region overlap, but all patches are still correlated through a common scan. There is no independent-specimen test of the trained model.

Simulated input generation

Each uint16 crop is mapped to an arbitrary attenuation scale by dividing by 65535 and multiplying by 0.4. The simulator uses 75 circular cone-beam views at 4.8-degree spacing and a 250-pixel detector in each dimension. Source-object and source-detector distances are borrowed from the geometry source. The 128-cubed simulation field is 12.000041 mm wide. This is an assumption for forward projection, not physical NMC calibration.

Matched comparison

FDK followed by 20 SIRT iterations gives the initial reconstruction. It is quantized through float16 and restored to float32 to match the model input. The raw model adds its predicted correction. The hybrid adds three SIRT steps after the model. The comparison baseline also adds three steps, producing SIRT23. This avoids attributing improvements solely to three extra physics iterations.

What a reference means here

The simulated reference is the input reconstruction crop. It allows an exact numeric comparison inside the simulator. It is still a reconstructed image, rather than an independent measurement of the true material. Matching forward and inverse geometry can make simulated results easier than measured-data reconstruction.

5 Residual 3D U Net and training

The compact network follows volumetric encoder-decoder ideas [4] and predicts a residual correction [5]. Its purpose is regression of reconstruction artifacts, not segmentation or classification. PyTorch supplies the learning framework [7].

Component	Implementation
Input and output	One channel; input plus predicted correction
Encoder feature widths	12, 24, 36
Convolution blocks	Two 3 x 3 x 3 convolutions with GELU
Downsampling	Stride-2 convolution
Upsampling	Trilinear interpolation plus skip concatenation
Output head	1 x 1 x 1 convolution; zero-initialized
Training patches	80 cubed; batch of two
Optimizer	AdamW; learning rate 0.0002; weight decay 0.0001
Loss	L1 voxel error plus 0.1 gradient L1 discrepancy
Training schedule	12 epochs; six updates per training region per epoch
Stability	Mixed precision; gradient norm clipped at 1
Selection	Fixed validation patches; epoch 0 eligible

The zero correction head makes the untrained model initially behave like the input reconstruction. It then learns corrections rather than having to reproduce the entire volume. Random XY right-angle rotations and Z flips augment matching input/target patches. Normalization divides both inputs and targets by 0.4.

Tiled inference

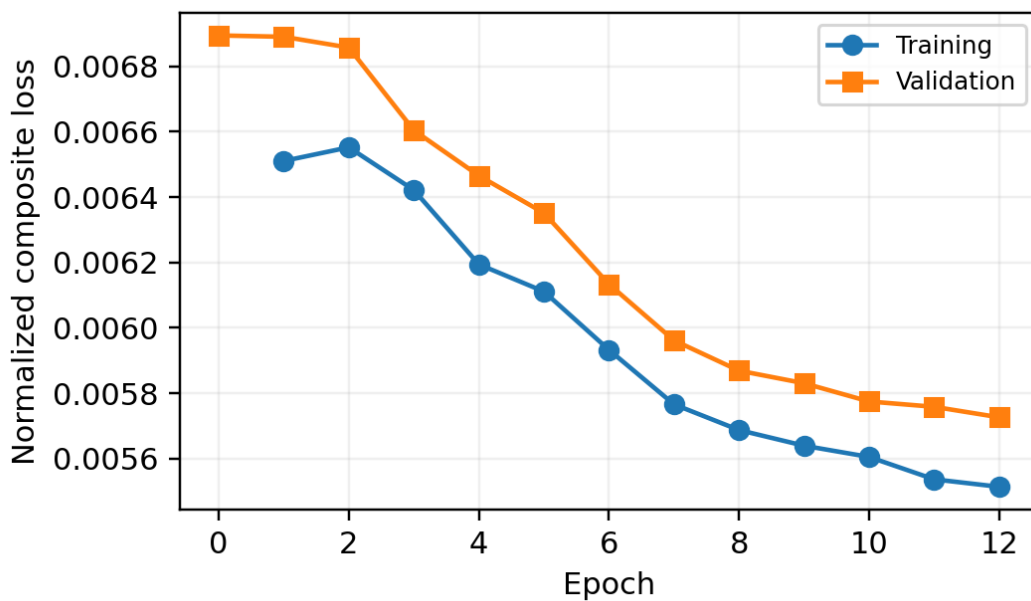
A full 128-cubed region uses 27 overlapping tiles: origins [0,40,48] along each axis, with 80-cubed tile size. A separable Hann window with a minimum weight of 0.05 per axis blends overlaps. Finite-output and coverage checks catch invalid predictions or uncovered voxels.

The selected epoch is 12. Validation composite loss decreases from 0.006895 to 0.005726; the latter is not the test voxel RMSE. The checkpoint is frozen before testing. Later test results must not be used to select a replacement checkpoint and still be called an untouched test.

6 NMC results and transfer failure

Method	0bar mean test RMSE	2000bar mean RMSE
sirt20	0.006373	0.006726
model_raw	0.004038	0.004406
sirt23	0.006155	0.006498
model_sirt3	0.003838	0.004199

On six 0bar test regions, the raw model gives 36.64% lower mean voxel RMSE than SIRT20. The hybrid gives 37.64% lower mean voxel RMSE than SIRT23. On six frozen 2000bar regions, the corresponding reductions are 34.49% and 35.38%, and the hybrid wins in all six regions. These are simulation results [10].



The learning curve records normalized composite loss on training and validation patches. It supports improvement during the planned 12 epochs. It does not establish optimal training duration or generalization beyond the scan and fixed simulation conditions.

The earlier frozen transfer failed

Before training this NMC model, a different epoch-8 battery checkpoint was applied to one fixed central 0bar crop. SIRT20 voxel RMSE was approximately 0.006662 while raw-model RMSE was 0.014350. The raw correction was 115.41% worse. After matched three-step SIRT corrections, the hybrid was 19.19% worse than SIRT23. That failed transfer led to the NMC-specific pilot. It must remain in the project history.

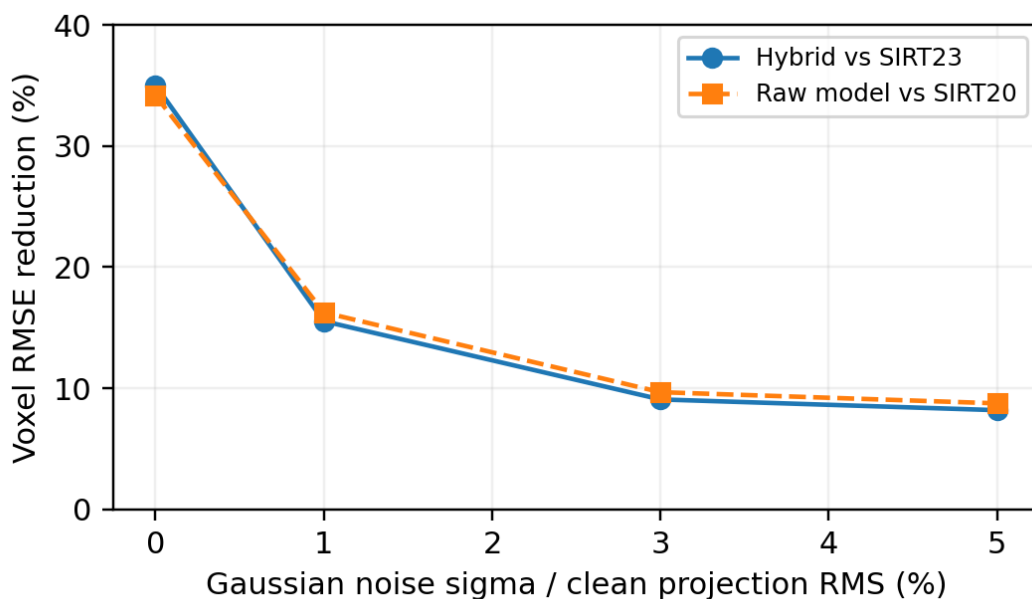
What can be claimed

The new model improves simulated reconstruction relative to the named controls on the tested regions. The 2000bar result shows separate-stack transfer under the same simulator. It does not establish different-specimen independence, real projection performance, battery defect accuracy, or a general-purpose reconstruction model.

7 Frozen noise sensitivity study

The epoch-12 model is fixed during this experiment. Six fresh regions from the 2000bar stack are evaluated at zero, 1%, 3% and 5% Gaussian noise. Noise is applied in projection line-integral units, with standard deviation equal to a fraction of the clean projection RMS. It is not a percentage of noisy pixels or a calibrated dose setting.

Noise fraction	Evaluations	Raw reduction	Hybrid reduction
0%	6	34.17%	35.03%
1%	18	16.26%	15.53%
3%	18	9.67%	9.08%
5%	18	8.75%	8.19%



There are six zero-noise evaluations and 18 at each nonzero level, totaling 60. Each nonzero level uses three seeds per region. Hybrid region-mean wins are 6/6 at every level. Repeated seeds characterize variation within those regions; they are not 60 independent specimens.

Interpretation

As noise increases, the hybrid gain falls from 35.03% to 8.19%. At nonzero noise the raw model has lower aggregate voxel RMSE than the hybrid, although comparisons use their respective matched physics baselines. This shows that improving the fit to noisy measurements can sacrifice fidelity to the clean simulated reference. Both metrics should therefore be reported.

Report serialization challenge

All 60 evaluations completed in an early run, but writing report.json failed because a region-win count was a NumPy int64. The fix converted the value to a Python int before JSON serialization. The completed rerun reproduced the numerical results. The failure occurred in report packaging, not training or reconstruction.

8 Measured Waygate preprocessing and geometry

The archived acquisition has 1200 used views spanning a complete rotation. The replay initially selects images 1, 17, 33, and so on through 1185, yielding 75 uniformly spaced angles. Image 1201 is excluded as the repeated endpoint. Selected timestamps still span 1184 s of an original 1199-s first-to-last acquisition.

Geometry item	Recorded or assumed value
Source-object distance	129.03973962 mm
Source-detector distance	806.495599 mm
Magnification	About 6.25
Exported projection shape	750 x 750
Reduced projection shape	250 x 250 after 3 x 3 averaging
Reduced detector pitch	0.6 mm, provisional
Vendor voxel size	About 0.03200011 mm
Replay field width	24.000082 mm
Model replay grid	128 cubed; voxel pitch about 0.1875006 mm
Physics diagnostic grid	256 cubed; voxel pitch about 0.0937503 mm

Provisional intensity transform

The implemented transform averages 3 x 3 intensity blocks, floors the mean at 0.5 and applies $-\log(I/9851)$. Slight negative results are retained. No extra dark/flat subtraction is applied because the correction status of exported TIFFs is unknown. The bright and dark file dimensions do not directly match the projections.

Unresolved calibration

Metadata ROI coordinates cover 1500 raw pixels but the export has 750 pixels. The effective detector pitch may depend on the vendor crop/binning convention. Vendor tilt and CorrectionValue are recorded but not interpreted. Scanner-to-ASTRA orientation is not independently verified. The first TIFF contains no sufficient correction-history proof. A visually plausible battery diameter is not a replacement for calibration.

The dataset authors describe these as acquired battery projections and vendor reconstructions [9]. The project uses a centered circular approximation, with all assumptions visible in its preparation report. This keeps exploratory replay reproducible without labeling it calibrated.

9 Measured NMC replay and negative transfer

Method	Fit projection RMSE
sirt20	0.057630
model_raw	0.062314
sirt23	0.055724
model_sirt3	0.058574

The lowest reported fit projection error is SIRT23 at 0.055724. The NMC hybrid gives 0.058574, an increase of 5.11%. The raw model is also worse than SIRT20. This result does not reproduce the model gains from NMC simulations.

Why the mismatch matters

The NMC network was trained on simulated electrode crops, while the Waygate archive depicts a cylindrical cell with layered structure and vendor-specific measurement processing. The initial replay also doubles voxel spacing relative to the training simulator. Texture, geometry, scale and correction status all change. The experiment cannot isolate one cause, but it establishes that this frozen transfer did not improve measured-data fit.

Pipeline timing

Stage	Recorded seconds
Total processing	3.112
Cold model loading and warmup	1.694
FDK plus SIRT20	0.447
Model inference	0.292
Extra baseline SIRT	0.095
Extra hybrid SIRT	0.095
Projection metrics	0.414

The total includes volume saving and other measured stages, but excludes scanner capture, archived projection preparation, browser delivery and ZIP packaging. It is not an acquisition-to-display measurement. PyTorch memory statistics exclude ASTRA allocations and therefore are not total device memory usage.

Separate earlier Waygate result

Earlier project logs reported reserved-view RMSE values of approximately 0.05382 for SIRT20, 0.05191 for SIRT23 and 0.05060 for a different model-plus-SIRT candidate, summarized as a 2.53% reduction. That checkpoint and protocol belong to a separate experiment. Its original complete review package is not revalidated by this report, and its result is not attached to the NMC checkpoint.

10 View count and resolution diagnostics

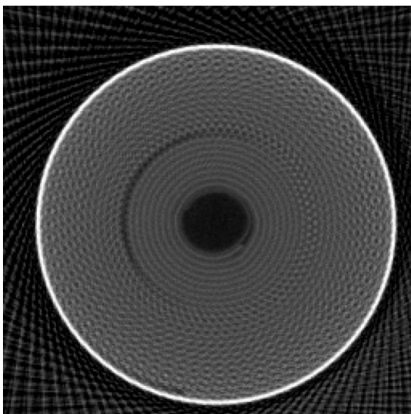
The controlled physics diagnostics use FDK alone, with the same provisional preprocessing and geometry. An initial comparison of 75 and 1200 views at 128 cubed and 256 cubed showed that both angular density and voxel resolution affect visible layers. All four runs completed successfully.

Views	Grid	Reconstruction time s
75	128 cubed	0.257
1200	128 cubed	0.223
75	256 cubed	0.209
1200	256 cubed	0.560

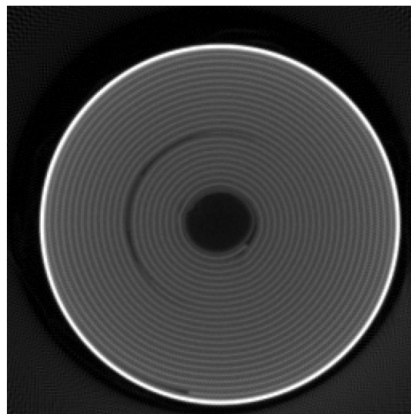
A second study fixes the grid at 256 cubed and compares nested subsets of 75, 150, 300, 600 and 1200 views. The 75- and 1200-view axial control arrays reproduce the first diagnostic exactly. The source-projection hash manifest is identical across the two runs.

Views	Reconstruction s	Axial RMSE vs 1200 views
75	0.211	0.023706
150	0.231	0.012385
300	0.271	0.003945
600	0.358	0.001060
1200	0.534	0.000000

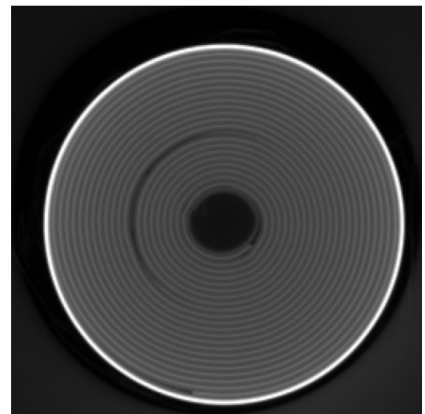
75 measured views



300 measured views



1200 measured views



All displayed FDK slices use the same brightness range in this illustration. The 75-view result has strong streaks; the 300-view result reveals layers with residual artifacts; 600 views is cleaner and closer to 1200 views. The vendor 750-pixel image is a separately processed reference, not a calibrated truth target.

The RMSE values above compare one axial slice with the same-scan full-view FDK slice. The 1200-view row is zero by definition. These values are not classification accuracy or a defect-preservation test. The small timing differences are single-run observations, including possible warmup effects, not a repeated performance benchmark.

11 Interfaces and presentation workflow

Local GPU inference app

The NMC local app runs on the GPU Python environment, checks dependencies, accepts supported inputs, and creates real outputs. It exposes simulation noise, model/physics comparisons, progress, cancellation and result exports. Simulation uses synthetic projections of reference crops; its measured mode requires compatible geometry and input assumptions.

Archived measured replay

The Waygate replay application reads the prepared measured projection array, records explicit provisional-calibration acknowledgments, and runs physics and frozen-model comparisons. Its successful execution verifies that the implementation can process that archive. It does not imply that the transferred model improves the reconstruction.

Standalone interactive review

waygate_interactive_review.html embeds actual completed view-count reconstructions. It provides five view-count choices, three saved middle directions, common brightness controls, a comparison overlay, PNG export, a numerical table, a downloadable JSON summary, and a plain-language explanation. It works offline and does not perform inference.

Demo mode	What it displays	What it does not establish
300-view Preview	Completed 300-view 256-cubed FDK	Live preview during acquisition
600-view Review	Completed cleaner 600-view FDK	Production quality threshold
1200-view reference	Same-scan full-view FDK comparison	Independent physical ground truth

How to demonstrate it

Open the standalone review in a browser. Start at 75 views so streaks are visible, switch to 300 and 600, then compare with 1200 using the common display scale. Change the slice direction to explain the reconstructed volume. Show the results page and clarify that the timings exclude capture. End with the remaining calibration and validation work.

A presenter should distinguish the live-compute app from the saved-results review. A slider or view-count button is an interface control, not evidence of streaming scanner operation.

12 Challenges and how they were handled

Challenge	Action taken	Remaining limit
Duplicate TIFF ZIP entries	Counted unique paths and verified slices	No extra specimens created
Large arrays and GPU work	Fixed crops, tiled inference and smaller detector grids	Full-resolution acquisition still costly
Scanner calibration mismatch	Preserved assumptions and source metadata	Effective pitch/correction unresolved
Frozen model failed transfer	Kept physics controls and negative results	Measured-domain benefit unproven
NumPy int64 JSON failure	Converted region count to Python int	Serialization must be checked
Missing ZIP/path commands	Used explicit downloads and quoted executable PowerShell commands	Files must exist locally
Misleading real-time interpretation	Separated solver and capture timing	No scanner-connected latency test

Scientific risks

The spatial split shares a scan and may learn scan-specific texture. Simulated forward and inverse operators match, which limits realism. Gaussian line-integral noise lacks a validated detector/photon model. Projection fit uses reconstruction angles in the NMC replay and is not a held-out test. All-view diagnostics now use previously reserved measurements, so subsequent work on them is development rather than fresh testing.

Reliable presentation

Percentage reductions must always name their baseline and metric. The 37.64% value is hybrid simulated voxel RMSE reduction relative to SIRT23. The latest Waygate 5.11% is an increase in fit projection RMSE. Sub-second FDK timings refer to reconstruction only. No publication result, model accuracy or specimen independence is inferred from interface polish.

Limits that remain

No battery defect labels, defect detection model, health predictor, motion-compensated 4D reconstruction, real-time scanner connection or independent-specimen performance study has been completed. Image plausibility cannot fill these gaps.

13 Real time roadmap and completion status

A real-time system must deliver useful information while the physical process is happening. Fast GPU reconstruction is necessary but insufficient. A static scan may still require time for exposure, rotation, transfer, preprocessing and display. The archived 1199-s scan is a concrete example.

Proposed next stages

Stage	Evidence needed to call it complete
Calibration	Documented crop/binning, pitch, gain/offset, center and orientation; reproducible checks
Acquisition integration	Authorized scanner feed; complete frames matched to correct logged angles
Latency measurement	Capture-to-display timings; stage breakdown; repeated cold/warm runs and backlog behavior
Task validation	Independent scans and defined structural/defect criteria
Measured-domain model	Training/development split separate from evaluation; matched physics and reserved-view controls
Deployment evaluation	Failure handling and operator interpretation tested in the intended workflow

These are future design requirements, not features implemented by the present reports. A replay can test reconstruction and interface behavior, but it cannot establish the real scanner throughput improvement that would follow from a sparse acquisition protocol.

What is complete now

The research demonstration includes data preparation, NMC training, fixed simulation tests, frozen transfer and noise studies, archived measured replay, view-count/resolution diagnostics, result packaging and interactive presentation. The important result is both the benefit in controlled NMC simulations and the observed limit of frozen transfer to measured Waygate data.

Appropriate project description

An interactive GPU-powered battery CT reconstruction prototype, evaluated with simulated NMC electrode projections and archived measured Waygate data, combining physics baselines, a residual 3D correction model, reproducible protocols, noise sensitivity studies and view-count diagnostics.

14 Reproducibility and evidence register

The exact experiment records should be retained with the code and data manifests. A percentage without its corresponding protocol, checkpoint and baseline is not a reproducible result [10]. The following records support this report.

Record	Evidence carried
NMC report.json and protocol.json	24/6/6 spatial split; locked epoch 12; test means
NMC training_history.json	Epoch 0 to 12 loss history
2000bar_frozen_check/report.json	Frozen model; six regions; separate-stack means
noise_robustness/report.json and locked_plan.json	60 evaluations; noise RMS rule; seeds; fixed regions
results (4).zip	Measured replay arrays and reported fit projection errors
preparation_report.json	75 selected views; angles; timestamps; hashes; assumptions
waygate_view_review.zip	75/1200 views at two grid sizes
waygate_view_count_review.zip	Five view counts; saved slices and times
PCA PCP PCR and middle reference TIFF	Acquisition, geometry, reconstruction and image context

Locked identities

NMC epoch-12 checkpoint SHA-256:

85d942f6d2697ab57a91c2e3204b0348

17e5555603183ef9bf90cfc7eaba8750

Training protocol SHA-256:

c180e08642f780d85ac121be74732bfc4

92735d9e88691e7a9ed3e9326b33c62

Prepared 75-view measured projection NPY SHA-256:

54bf2d838b45004eddea855c59bcd9ddc

23891692546b42e90998277796a3670

Verification scope

Uploaded ZIP integrity and finite saved arrays were checked. View-count axial RMSE values were independently recalculated from the saved slices. The 75/1200-view axial controls were byte-identical numerically across diagnostics. The measured replay fit errors are reported by its GPU run; they were not independently reprojected in the report-authoring environment. Timing values are recorded observations, not independent benchmark replications.

Earlier dynamic 41.2% and synthetic 2.63% claims are not carried into this report as established results, because their complete source records and comparison definitions are not part of the present verified experiment set. They require a separate evidence review before publication or resume use.

15 References and citation guide

Bracketed citations identify external methods, data sources and the unpublished experiment record. External literature supports design context, not the numerical results of this project. Source links were checked on 4 October 2026. The companion paper follows an IEEE-style two-column layout and numbered reference convention; specific venue submission requirements should be checked against the official template [11].

- [1] L. A. Feldkamp, L. C. Davis, and J. W. Kress, "Practical cone-beam algorithm," *Journal of the Optical Society of America A*, vol. 1, no. 6, pp. 612-619, 1984. doi: 10.1364/JOSAA.1.000612. [Open source](#).
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- [9] E. Zwanenburg and J. Warnett, "XCT datasets of battery for reconstruction," Zenodo, version v1, 2025. doi: 10.5281/zenodo.14993742. [Open source](#).
- [10] A. P. K. Goud, "Sparse4D battery extension experiment records," unpublished project archive, Oct. 4, 2026. Includes protocols, checkpoint hashes, training and noise reports, archived replay results, and view-count diagnostics.
- [11] IEEE Author Center, "Authoring tools and templates," conference author resources. Accessed Oct. 4, 2026. [Open source](#).

For academic reuse, cite the Kaggle and Zenodo datasets when describing the data, the FDK and ASTRA sources when describing reconstruction, and the architecture sources when explaining inspiration. Treat project results as original unpublished experimental observations until reviewed and published.